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Use of tyrosine kinase inhibitors for pediatric philadelphia chromosome-positive acute lymphoblastic leukemia: a systematic review and meta-analysis.

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Use of tyrosine kinase inhibitors for pediatric philadelphia chromosome-positive acute lymphoblastic leukemia: a systematic review and meta-analysis.

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ABSTRACT

Objectives Tyrosine kinase inhibitors (TKIs) are regarded as a promising therapeutic approach for pediatric Philadelphia chromosome-positive acute lymphoblastic leukemia (Ph+ALL). The objective of this review is to systematically investigate the effectiveness and safety of TKIs in the management of pediatric Ph+ALL.

Methods Electronic searches were conducted in CENTRAL, MEDLINE, EMBASE, SIOP, ASPHO, ASCO, ASH and four Chinese databases. Prospective or retrospective comparative studies were included. Two authors independently assessed the full text and extracted data. Quality of studies was assessed by the Cochrane Collaboration's risk of bias tool and Newcastle-Ottawa quality assessment scale. Subgroup analysis was performed by different types of TKIs.

Results Two randomized controlled trials and four cohort studies enrolling 536 patients were included. The quality of these references was moderate. For trials, the pooled hazard ratio (HR) was 0.68(95%CI, 0.26-1.78) in overall survival (OS), 0.63(95%CI: 0.28-1.42) in event-free survival (EFS), respectively, comparing TKIs arm with non-TKIs arm for treatment of pediatric ph+ALL. There was significant difference in OS and EFS between dasatinib arm and imatinib arm (HR: 2.26, 95%CI: 1.02-5.01; HR: 2.36; 95%CI: 1.27-4.39, respectively). For cohort studies, the pooled HR was 0.25(95%CI: 0.14-0.47) in OS, 0.25(95% CI: 0.14-0.44) in EFS, respectively, comparing TKIs arm with non-TKIs arm. There were no significance differences in adverse drug reaction between TKI or without TKI treatment (RR: 0.82, 95%CI: 0.63-1.08 in RCT; RR: 1.01, 95%CI: 0.64-1.59 in cohort studies; respectively), and dasatinib versus imatinib (RR: 0.97, 95%CI: 0.77-1.23).

Conclusions Medium-quality studies suggest that the combination of early and continuous exposure to TKIs and intensive chemotherapy are likely to increase the OS

and EFS rates for pediatric Ph+ALL. The second-generation TKI dasatinib is superior to the first-generation imatinib in the OS and EFS. The adverse drug reactions of TKIs could be tolerated.

Keywords: tyrosine kinase inhibitor; Ph+ALL; pediatric

PROSPERO registration number CRD42018104107

Strengths and limitations of this study

This study is the first systematic review of the effectiveness and safety of TKIs in the management of pediatric Ph+ALL.

Our findings reveal that TKIs, especially the second-generation TKI combined with intensive chemotherapy may be beneficial for the treatment of pediatric Ph + ALL.

This is a meta-analysis of prospective and retrospective comparative studies.

Introduction

Philadelphia chromosome-positive acute lymphoblastic leukemia (Ph+ALL), which occurs in approximately 3-5% of pediatric ALL patients, is recognized as a severe disease but leading to a dismal prognosis than those with philadelphia chromosome-negative ALL.[1]

In the past decades, intensive chemotherapy was used in the treatment of pediatric Ph+ALL patients. [2,3] Afterwards, hematopoietic stem cell transplant (HSCT) in first remission was considered the treatment of choice, due to benefit rate of overall survival (OS) and event-free survival (EFS) than intensive chemotherapy alone. However, the mortality rate associated with transplantation is almost half. [4,5] Recent data indicated that tyrosine kinase inhibitors (TKIs) targeting BCR-ABL fusion protein combined with chemotherapy maybe an alternative effective therapy. [6] In the Ph+ALL adult and young patients' practice, TKIs in combination with chemotherapy were capable to increase the complete remission and HSCT rates and improved the early outcome. [7] In pediatric patients, the Children's Oncology Group (COG) trial AALL0031 had shown 80% 3-year EFS for Ph+ALL patients treated with intensive chemotherapy plus continuous 340mg/m^2 imatinib. The addition of imatinib to chemotherapy has similar outcomes to HSCT especially in patients who respond favorably. [8] Longer-term follow-up demonstrated some attrition in survival but confirmed the overall favorable outcome and HSCT does not appear a benefit. [9] However, the trial was observational and confirmatory results are not yet available

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from the European intergroup study on post induction treatment of ph+ALL with imatinib trial. [10]

Newer TKIs including dasatinib and nilotinib, were developed for those patients with resistance to imatinib. The resistance mechanisms are commonly associated with acquiring kinase domain mutations, reduced drug availability and activation of other signaling pathways, such as the SRC family kinases. [11,12] Dasatinib is the more commonly used dual ABL/SRC kinase inhibitor and can cross the blood-brain barrier to eradicate central nervous system (CNS leukemia), and activity in most cases of imatinib resistance. [13,14] For patients with BCR-ABL T315I mutation ph+ALL, the third-generation TKI ponatinib was reported to become a valuable alternative treatment option. [15,16]

Incorporation of TKIs into frontline regimens for patients with newly diagnosed Ph+ ALL may be the current general consensus. However, this view may not be universally accepted so it will be helpful to analyze systematically to see if there is a true benefit to the addition of TKIs to conventional chemotherapy. Therefore, we object to investigate the current available evidence on the effectiveness and safety of TKIs in the management of pediatric Ph+ALL.

Methods

This protocol is registered with the international PROSPERO register of systematic reviews: CRD42018104107.

Patient and Public Involvement

Patients and the public were not be involved in this review.

Search strategy

We searched the Cochrane Central Register of Controlled Trials (CENTRAL) published in the Cochrane Library (latest issue), MEDLINE in PubMed (from 1966 to present) and EMBASE (Ovid; from 1980 to present) using the search strategies

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detailed in Appendix 1. We searched the Chinese Biomedical Literature Database (CBM), Chinese National Knowledge Infrastructure (CNKI), VIP Database for Chinese Technical Periodicals (VIP) and WANFANG using the search strategies in Appendix 2. Language of publications was restricted in English and Chinese.

The following societies of conference proceedings of annual meetings (from the inception of TKIs treatment till now) were scanned：

1. International Society for Paediatric Oncology (SIOP);
2. American Society for Pediatric Hematology and Oncology (ASPHO);
3. American Society of Clinical Oncology (ASCO);
4. American Society of Hematology (ASH).

Inclusion and exclusion criteria

Study designs

There were likely scarce randomized controlled trials in the treatment of pediatric patients with Ph+ALL, hence, we included retrospective or prospective cohort/case-control/randomized studies. Studies that did not include a comparative group of patients, duplicate studies, abstracts and historical control studies were excluded.

Participants

ALL patients aged 1-18 years who were diagnosed by cytogenetic or molecular Ph positive status were included. Participants with ph-negative at diagnosis but positive at relapse, more than two types of malignant neoplasm, relapse or refractory Ph+ALL were excluded.

Interventions

TKIs alone, TKIs combined with other intervention

Comparators

Treatment without TKIs

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Outcomes

1. Overall survival (OS), defined as time from start of treatment to death from any cause;

2. Event-free survival (EFS), defined as time from start of treatment to an event, such as remission failure, relapse, death from any cause, abandonment of treatment, or second malignancy, or to the date of last follow-up contact;

3. Adverse drug reaction (ADR)， collected and scored according to the National Cancer Institute Common Toxicity Criteria for Adverse Events (NCI-CTCAE) or other criterion by investigators and clinical research coordinators.

Data extraction

Literature search results were uploaded to EndNote X7. Two authors independently screened the title and abstract of all studies identified by the search strategy and obtained full articles for all potentially relevant trials. Any disagreements were resolved by consensus between authors or achieved final resolution using a third-party arbitrator.

Two authors independently used a pre-designed data collection form to extract data from each study. The items included: characteristics of the trial (author, country, publication year, design etc), participants (age, sex etc), interventions (dose, regimen etc), outcomes and length of follow-up.

Risk of bias in individual studies

Two authors independently assessed the risk of bias using a standard form. For randomized trials, we used the domain-based evaluation recommended by the Cochrane Handbook for Systematic Reviews of Intervention to address the following domains: bias arising from the randomization process, bias due to deviations from intended interventions, bias due to missing outcome data, bias in measurement of the outcome, bias in selection of the reported result. [17] Plots of "Risk of bias" assessment were created using Review Manager 5 (RevMan 2014). For non-randomized trials, we used the Newcastle-Ottawa Scale (NOS) items which categorized into three broad

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perspectives: the selection of the study groups, the comparability of the groups, and the ascertainment of either the exposure or outcome of interest for case-control or cohort studies respectively. [18] The total score of NOS is 9 stars.

Data analysis and synthesis

Strategy for data analysis

For dichotomous data, we used the risk ratio (RR) with 95% confidence interval (95% CI) as the effect measure. For time-to-event data, which take into account of the number and timing of events, we summarized and analyzed data using hazard ratio (HR) with their corresponding 95% CI. [17] We used Parmar's method if HR did not been explicitly presented in the study. [19]

Assessment of heterogeneity

We quantified heterogeneity using the I^2 statistic, which illustrated the percentage of the variability in effect estimates resulting from heterogeneity. In the absence of significant heterogeneity (I^2 less than 50%), we used a fixed-effect model for the estimation of treatment effect. [17] Otherwise, we explored possible reasons for the occurrence of heterogeneity and used random-effects model.

Data synthesis

We entered data into the Review Manager 5 software and made analyses according to the guidance provided in the *Cochrane Handbook for Systematic Reviews of Interventions*. [20] We pooled results only if both treatment groups were comparable, including the outcome definition. If not, we presented a narrative summary.

Assessment of reporting biases

If we include 10 or more trials, we will use funnel plots to assess biases including publication bias and other reporting biases. If there are biases, the funnel plots may be asymmetrical.

Subgroup analysis

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Considering that different types of TKIs might affect the outcomes, we conducted a subgroup analysis.

Results

Study Selection

Our search strategy yielded 2476 references from the electronic searches and 69 from other sources. 252 duplicate publications and 1928 articles were excluded based on title and abstract for certainly not fulfilling inclusion criteria on the basis of the type of article, study design, population or outcome of interest. 365 articles underwent full-text review and six literatures were included in the data analysis. The PRISMA flow chart of bibliographic search was illustrated in Figure 1.

Characteristics of Included Studies

Six articles [10,21-25] (two prospective randomized open-label controlled trials and four retrospective cohort studies) were included. Five studies [10,22-25] assessed the efficacy and safety of TKIs in association with multidrug chemotherapy for pediatric patients with Ph+ALL. One randomized clinical trial [21]compared the effect of dasatinib versus imatinib. 536 children aged 1 to 18 years with a confirmed diagnosis of ph+ ALL were contributed to the meta-analysis. The chemotherapy regimen was modeled on Berlin-Frankfurt-Munster high-risk arm, the CCLG-ALL high-risk arm and the Chinese Children's Cancer Group study ALL (CCCG-ALL) intermediate-risk arm. TKI drugs include first generation imatinib or second generation dasatinib. Patients were prescribed oral imatinib (260-340mg/m^2.d) or dasatinib (40-80mg/m^2.d) on day 8, day 22 of remission induction or the end of induction. The exposure of TKIs ranged from 56 days to as long as continuing until the end of therapy. Table 1 described the detailed characteristics of the studies included in this meta-analysis.

Risk of Bias Assessment in Included Studies

Figure 2 provided quality assessment of randomized clinical trial and Table 2 described quality assessment of non-randomized clinical trial. In two prospective trials, stratified randomization was done centrally with web-based randomization system and both of them were open-label. Therefore, the random sequence generation was low risk while the blindness was high risk. All studies had low risk bias in incomplete outcome data, selective reporting and other issues. In four cohort studies,

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the NOS methodological score were 7-8 stars. In the selection section, both exposed cohort and unexposed cohort were drawn from the same source of participant. Data was collected and retrospective reviewed from medical records. Outcome was not present at the beginning of all studies. In the comparability section, the three studies did not describe important confounding factors, except for TKI treatment between exposed and unexposed cohort among pediatric ph+ALL. In the outcome section, one study was followed up for 2 years and may not be long enough for outcomes to occur. One study did not describe the patient's loss of follow-up. The retrospective research has high quality NOS scores. Overall, the risk of bias of RCTs was rated as moderate and the cohort studies were low.

Overall Survival

TKI versus non-TKI

Four studies (one RCT, three cohort studies) [10,22,23,25] with a total of 294 patients were included in the meta-analysis of OS for pediatric ph+ALL patients. The pooled HR of OS in patients with TKI use was 0.68 (95% CI: 0.26-1.78, *P*=0.43) in RCT and 0.25 (95% CI: 0.14-0.47, *P*<0.05) in cohort studies, respectively. Figure 3 shows the forest plot of the included studies.

Different TKIs

One trial [21] with a total of 189 Ph+ALL children compared the OS of oral imatinib at a daily dosage of 300mg/m^2 versus dasatinib at a daily dosage of 80mg/m^2, in combination with an intensive chemotherapy backbone. The 4-year OS rate in the dasatinib group was significantly higher than the imatinib group (88.4% versus 69.2%; *P*=0.04). The HR of OS was 2.26 (95%CI: 1.02-5.01, *P*=0.04). Figure 4 shows the forest plot of the study.

Event-free Survival

TKI versus non-TKI

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The primary outcome in one RCT [10] was disease-free survival (DFS) because the group only included good-risk patients in first complete remission. Poor-risk patients were not included, so DFS is equivalent to EFS in meta-analysis. Four studies (one RCT, three cohort studies) [10,23-25] with a total of 253 patients were included. The pooled HR of EFS in ph+ALL patients with TKI use was 0.63 (95%CI: 0.28-1.42, *P*=0.26) in RCT and 0.25 (95%CI: 0.14-0.44, *P*<0.05) in cohort studies, respectively. Figure 5 shows the forest plot of the included studies.

Different TKIs

One trial [21] with a total of 189 patients was included. In this trial, the 4-year EFS rate for the 92 dasatinib-treated patients was 71.0%, significantly better than the 48.9% for the 97 imatinib-treated patients (*P*=0.005).The HR of EFS was 2.36 (95%CI: 1.27-4.39; *P*<0.05). Figure 6 shows the forest plot of the study.

Adverse drug reaction

TKI versus non-TKI

Three studies (one RCT, two cohort studies) [10,22,24] reported serious ADR in patients with ph+ALL. Heterogeneity in the cohort studies was significant (I^2 =58%), then we used a random-effect model for the estimation of treatment effect. In RCT, proportions of patients with the most commonly reported serious ADR did not differ substantially between imatinib group and non-imatinib group (*P*=0.64), and were mainly caused by myelosuppression. The most common serious ADR was infection, including fungal infection, localized infection and others.The pooled RR of serious ADR in patients with TKI use was 0.82 (95%CI: 0.63-1.08, *P*=0.16) in RCT. Result of the cohort studies was consistent with RCT. The pooled RR was 1.01 (95%CI: 0.64-1.59, *P*=0.12) (Figure 7).

Different TKIs

One trial [21] reported the comparison of severe toxicity between patients treated with imatinib or dasatinib. There was no significant differences in the frequency of severe toxic effects between dasatinib and imatinib arms (RR: 0.97; 95%CI: 0.77-1.23, *P*=0.81) (Figure 8). The most frequently reported effects were grade 5 (fatal)

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infection, grade 3/4 infection, disseminated fungal infection, pancreatitis, seizure, thrombosis, intestinal hemorrhage, pleural effusion and hyperbilirubinemia. Infections constituted the most common serious adverse events, and approximately 5% of the patients in each arm died of fatal infections. Moreover, approximately 7% of patients in each treatment arm had disseminated fungal infections.The incidence of pleural effusion was 2% in the imatinib group and 4% in the dasatinib group. However, this difference was also not significant (P=0.44).

Discussion

TKIs have revolutionized the management of ph+ALL patients, but it is unclear whether this observed improvement is significant. This is the first systematic review to investigate whether adding TKIs to conventional chemotherapy is indeed beneficial. Eventually, a meta-analysis was performed for one RCT and four non-RCT with or without TKI treatment, and one RCT comparing dasatinib and imatinib in pediatric ph+ALL. The quality of these studies is medium.

TKI versus non-TKI

The included randomized trial EsPhALL 2004[10] demonstrated that imatinib in conjunction with intensive chemotherapy might be beneficial for treatment of children with ph+ALL. The 4-year DFS in good-risk patients who received intensive chemotherapy and discontinuous post induction imatinib 300mg/m2.d increased by approximately 20% compared with patients receiving chemotherapy alone (75.2% versus 55.9%). The long-term follow up outcome showed 5-year DFS in the imatinib arm of 75.5% versus 61.4% in the no imatinib arm. [26] Nevertheless, the combination use of TKIs and intensive chemotherapy didn't significantly improve OS and EFS, compared to those who did not use TKIs. The pooled HR was 0.68 (95% CI: 0.26-1.78) in OS, 0.63 (95% CI: 0.28-1.42) in EFS, respectively, comparing TKIs arm with non-TKIs arm. The results were reversed by the meta-analysis including four high-quality cohort studies (NOS score ≥7). In these cohort studies， the pooled HR

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was 0.25 (95% CI: 0.14-0.47) in OS, 0.25 (95% CI: 0.14-0.44) in EFS, respectively, comparing TKIs arm with non-TKIs arm. The reasons for the different results may be caused by the different timing and dosage of TKIs and the interference of HSCT during treatment.

The first reason to consider is that the use of imatinib in the study EsPhALL 2004 was induced at a dose of 300mg/m^2 for 126 days after induction. In cohort studies, imatinib 260-340mg/m^2 were started after induction or earlier on day 15 of remission induction. Most studies have continued to use TKIs throughout chemotherapy unless severe infections, intolerance to TKIs or neutropenia with fever. Early and continuous exposure of imatinib may improve outcomes of OS and EFS. However, a prospective, open-label, single-arm clinical trial EsPhALL2010, which is the successor of EsPhALL 2004, administrated imatinib 300mg/m2.d continuously and from day 15 of induction, displayed similar results to EsPhALL2004. [27] Based on the experience of other single-arm COG study [89] and the results of this meta-analysis, we tend to support the use of increased exposure to TKIs.

The second reason to consider is the interference of HSCT. There are limited data on outcome comparison between TKIs and chemotherapy versus HSCT treatment. The investigators from the COG reported the long-term follow-up of their study in ph+ALL children showing equivalent outcomes in patients who received at least 280 continuous days of imatinib to those who underwent HSCT. [9] Two other retrospective studies indicated that there was no significant difference in OS or EFS between HSCT and chemotherapy plus TKIs treatment, which was consistent with COG study. [28,29] In the included study EsPhALL 2004[10], about 80% of the patients in the study had HSCT, 37 out of 46 in the imatinib group and 32 out of 44 in the non-imatinib group, respectively. This might lead to no statistical difference between the two groups of results in this RCT. However, between TKI combined chemotherapy and HSCT treatment, the outcomes of pediatric ph+ALL patients has not been proven.

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Different TKIs

In previous studies of imatinib, EFS was approximately 60% at 5 years, but resistance and relapse is common. Disappointingly, COG trial (AALL0622) for treatment of pediatric Ph+ ALL did not prove the superior of dasatinib over imatinib. OS and EFS were similar to those in the preceding trials using imatinib. 30 Another COG trial (AALL1122) using dasatinib 60mg/m^2.d is ongoing. [31] Now, Shen et al [21] reported different outcomes from the first randomized trial directly comparing dasatinib with imatinib. In our meta-analysis of these 189 eligible patients, there was significant difference in OS and EFS between dasatinib arm and imatinib arm (HR: 2.26, 95%CI: 1.02-5.01; HR: 2.36, 95%CI: 1.27-4.39; respectively). It is worth noting that the dose of dasatinib is higher than COG trials and longer follow-up is required to assess whether they represent significant improvement.

Adverse drug reaction

The dosage of imatinib (260-570mg/m^2.d) and dasatinib (50-110 mg/m^2.d) are well tolerated in children and adolescents with leukemia. [32,33] In the included studies, the daily dose of imatinib and dasatinib were 260-340mg/m^2 and 40-80mg/m^2, suggesting that imatinib or dasatinib combined with intensive chemotherapy was well tolerated. The use of TKI did not increase the chemotherapy-related toxicity and side effects such as cytopenia and pleural effusions were mostly moderate and manageable. [34] There were no significance differences in severe adverse drug reaction between TKI or without TKI treatment (RR: 0.82, 95%CI: 0.63-1.08 in RCT; RR: 1.01, 95%CI: 0.64-1.59 in cohort studies; respectively). Similarly, no significant difference in the frequency of severe ADR was detected between dasatinib and imatinib arm (RR: 0.97, 95%CI: 0.77-1.23).

Limitations

There are some limitations. Firstly, it must be emphasized that all these analyses are limited by the small number of patients. In addition, HSCT is also very important

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factor that need to take into consideration. The addition of the TKIs had improved the 5-year EFS and OS rates, but virtually about 40% to 80% patients were transplanted. Lastly, this is a meta-analysis of prospective and retrospective studies with its inherent limitations. Therefore, we did not combine the results of different types of studies.

Conclusions

Medium-quality studies suggest that TKIs combined with intensive chemotherapy are likely to enhance the OS and EFS rates in pediatric Ph+ALL. The second-generation TKI dasatinib is superior to the first-generation imatinib in the OS and EFS. The adverse drug reactions of TKIs could be tolerated.

Further prospective controlled studies are warranted to address remaining questions relating to the appropriate timing for use of TKIs during chemotherapy, as well as the role of HSCT, in the management of pediatric ph+ALL.

Data sharing statement

No additional data are available.

Ethics and dissemination

The result of this systematic review will be disseminated by publication in a peer-reviewed journal. Ethical assessment was not needed because we searched and evaluated only existing sources of literature. Our findings should be beneficial to health professionals who prescribe TKIs for pediatric ph+ALL.

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Contributors MC and LLZ contributed to the conception of the study. The

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manuscript of the review was drafted by MC and revised by YPZ. YZL and TZTW independently screened the potential studies and extracted data from the included studies. YZL and TZTW assessed the risk of bias and finished data synthesis. LLZ arbitrated any disagreements and ensured that no errors occur during the review. All authors read, provided feedback and approved the final manuscript.

Competing interests None declared.

References

1. Schlieben S, Borkhardt A, Reinisch I, *et al*. Incidence and clinical outcome of children with BCR/ABL-positive acute lymphoblastic leukemia (ALL). A prospective RT-PCR study based on 673 patients enrolled in the German pediatric multicenter therapy trials ALL-BFM-90 and CoALL-05-92. *Leukemia* 1996;10(6):957-63.
2. Reiter A, Schrappe M, Ludwig WD, *et al*. Chemotherapy in 998 unselected childhood acute lymphoblastic leukemia patients. Results and conclusions of the multicenter trial ALL-BFM 86. *Blood* 1994;84(9):3122-33.
3. Ribeiro RC, Broniscer A, Rivera GK, *et al*. Philadelphia chromosome-positive acute lymphoblastic leukemia in children: durable responses to chemotherapy associated with low initial white blood cell counts. Leukemia 1997;11(9):1493-6.
4. Aricò M, Schrappe M, Hunger SP, *et al*. Clinical outcome of children with newly diagnosed Philadelphia chromosome-positive acute lymphoblastic leukemia treated between 1995 and 2005. *Journal of clinical oncology* 2010;28(31):4755-61.
5. Arico M, Valsecchi MG, Camitta B, *et al*. Outcome of treatment in children with Philadelphia chromosome-positive acute lymphoblastic leukemia. *N Engl J Med* 2000;342(14):998-1006.
6. Ravandi F. How I treat Philadelphia chromosome-positive acute lymphoblastic leukemia. *Blood* 2019;133(2):130-6.
7. NCCN. Clinical Practice Guidelines in Oncology. Acute Lymphoblastic Leukemia. Secondary Clinical Practice Guidelines in Oncology. Acute Lymphoblastic Leukemia 2018. https://www.nccn.org/professionals/physician_gls/default.aspx (accessed 29 Aug 2019).
8. Schultz KR, Bowman WP, Aledo A, *et al*. Improved early event-free survival with imatinib in Philadelphia chromosome-Positive acute lymphoblastic leukemia: A Children's Oncology Group Study. *Journal of Clinical Oncology* 2009;27(31):5175-81.
9. Schultz KR, Carroll A, Heerema NA, *et al*. Long-term follow-up of imatinib in pediatric Philadelphia chromosome-positive acute lymphoblastic leukemia: Children's oncology group study AALL0031. *Leukemia* 2014;28(7):1467-71.
10. Biondi A, Schrappe M, De Lorenzo P, *et al*. Imatinib after induction for treatment of children and adolescents with Philadelphia-chromosome-positive acute

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lymphoblastic leukaemia (EsPhALL): a randomised, open-label, intergroup study. *The lancet oncology* 2012;13(9):936-45.
11. Brave M, Goodman V, Kaminskas E, *et al*. Sprycel for chronic myeloid leukemia and Philadelphia chromosome-positive acute lymphoblastic leukemia resistant to or intolerant of imatinib mesylate. *Clin Cancer Res* 2008;14(2):352-9.
12. Kantarjian H, Shah NP, Hochhaus A, *et al*. Dasatinib versus imatinib in newly diagnosed chronic-phase chronic myeloid leukemia. *N Engl J Med* 2010;362(24):2260-70.
13. O'Hare T, Walters DK, Stoffregen EP, *et al*. In vitro activity of Bcr-Abl inhibitors AMN107 and BMS-354825 against clinically relevant imatinib-resistant Abl kinase domain mutants. *Cancer Res* 2005;65(11):4500-5.
14. Porkka K, Koskenvesa P, Lundan T, *et al*. Dasatinib crosses the blood-brain barrier and is an efficient therapy for central nervous system Philadelphia chromosome-positive leukemia. *Blood* 2008;112(4):1005-12.
15. Cortes JE, Kantarjian H, Shah NP, *et al*. Ponatinib in refractory Philadelphia chromosome-positive leukemias. *N Engl J Med* 2012;367(22):2075-88.
16. Nicolini FE, Basak GW, Kim DW, *et al*. Overall survival with ponatinib versus allogeneic stem cell transplantation in Philadelphia chromosome-positive leukemias with the T315I mutation. *Cancer* 2017;123(15):2875-80.
17. Higgins J GS. Cochrane handbook for systematic reviews of interventions. . Chichester, UK: The Cochrane Library, John Wiley &Sons https://methodscochraneorg/risk-bias-20-tool (accessed 25 Jan 2019) 2018.
18. Wells GA SB, O'Connell D, *et al*. The Newcastle-Ottawa Scale (NOS) for assessing the quality if nonrandomized studies in meta-analyses. http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp (accessed 29 Aug 2019).
19. Parmar MK, Torri V, Stewart L. Extracting summary statistics to perform meta-analyses of the published literature for survival endpoints. *Stat Med* 1998;17(24):2815-34.
20. Review Manager(RevMan)[Computer program] 2014. Version 5.3.Copenhagen: The Nordic Cochrane Centre, The Cochrane Collaboration.
21. Shen Sh, Chen XJ, Cai J, *et al*. Effect of Dasatinib vs Imatinib in the Treatment of Pediatric Philadelphia Chromosome-Positive Acute Lymphoblastic Leukemia: A Randomized Clinical Trial. *JAMA Oncology* 2020;6(3):358-66.
22. Zhang J, Li ZF, Guo J *et al*. Clinical Observation of Inductive Treatment of Childhood Acute Lymphoblastic Leukemia with Imatinib Mesylate Combined with VDLD R egimen. *Clinical Misdiagnosis & Mistherapy* 2019;32(9):26-30.
23. Wang CC, Jin RM. Clinical Analysis of 19 BCR-ABL Positive Childhood Acute Lymphoblastic Leukemia. Master dissertation. Huazhong University of Science and Technology. 2016
24. Guo Y, Liu TF, Ruan M *et al*. Efficacy and safety of imatinib for the treatment of Philadelphia chromosomepositive acute lymphoblastic leukemia in children. *Chin J Contemp Pediatr* 2015;17(8):819-24.
25. Liu TF, Zhu XF. Clinical and biological characteristics, therapeutic effects and

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prognostic factors of 92 children with Ph chromosome positive acute lymphoblastic leukemia. Doctor dissertation. Peking Union Medical College. 2013.

26. Biondi A, Cario G, De Lorenzo P, *et al*. Long-term follow up of pediatric Philadelphia positive acute lymphoblastic leukemia treated with the EsPhALL2004 study: high white blood cell count at diagnosis is the strongest prognostic factor. *Haematologica* 2019;104(1):e13.
27. Biondi A, Gandemer V, De Lorenzo P, *et al*. Imatinib treatment of paediatric Philadelphia chromosome-positive acute lymphoblastic leukaemia (EsPhALL2010): a prospective, intergroup, open-label, single-arm clinical trial. *Lancet Haematology* 2018;5(12):641-52.
28. Salami K, Alkayed K, Halalsheh H, *et al*. Hematopoietic stem cell transplant versus chemotherapy plus tyrosine kinase inhibitor in the treatment of pediatric Philadelphia chromosome-positive acute lymphoblastic leukemia (ALL). *Hematology/ Oncology and Stem Cell Therapy* 2013;6(1):34-41.
29. Gao Y, Guo Y, Hu SY, *et al*. Philadelphia chromosome-positive acute lymphoblastic leukemia in China: a retrospective study from the Chinese Childhood Cancer Group. *Leukemia & lymphoma* 2016;57(11):2696-8.
30. Slayton WB, Schultz KR, Kairalla JA, *et al*. Dasatinib Plus Intensive Chemotherapy in Children, Adolescents, and Young Adults With Philadelphia Chromosome-Positive Acute Lymphoblastic Leukemia: Results of Children's Oncology Group Trial AALL0622. *Journal of clinical oncology* 2018;36(22):2306-14.
31. ClinicalTrials.gov. Pediatric philadelphia positive acute lymphoblastic leukemia. Available at: https://www.clinicaltrials.gov/ct2/show/NCT01460160?term=AALL1122&draw=2&rank=1. Accessed July 10.2020.
32. Champagne MA, Capdeville R, Krailo M, *et al*. Imatinib mesylate (STI571) for treatment of children with Philadelphia chromosome-positive leukemia: results from a Children's Oncology Group phase 1 study. *Blood* 2004;104(9):2655-60.
33. Aplenc R, Blaney SM, Strauss LC, *et al*. Pediatric phase I trial and pharmacokinetic study of dasatinib: a report from the children's oncology group phase I consortium. *Journal of clinical oncology* 2011;29(7):839-44.
34. Lindauer M HA. Dasatinib. *Recent Results Cancer Res* 2018;212:29-68.

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Table 1. Main characteristics of the studies included in this meta-analysis.

Study/Year	Setting	Time	Population (years)	TKI start time	TKI group n1	intervention	Non-TKI group n2	intervention	outcome
Randomized controlled trials									
1 Biondi 2012 [10]	Europe	2004.1-2009.12	1-18	post-induction, good risk	46	Imatinib(300mg/m^2.d)+ Chemotherapy	44	Chemotherapy	OS,EFS, ADR
2 Shen 2020 [21]	China	2015.1-2018.9	7.8(0-18)	After diagnosis, usually on D8 of remission induction	97	Imatinib(300mg/m^2.d)+ Chemotherapy	92	Dasatinib(80 mg/m^2.d)+ Chemotherapy	OS,EFS
Cohort studies									
1 Zhang 2019 [22]	China	2008.1-2012.12	8.26(2-12)	post-induction	54	Imatinib(0.6g/d)+ Chemotherapy	40	Chemotherapy	OS, ADR
2 Wang 2016 [23]	China	2008.10-2016.1	9(3-15)	Post-induction	9	Imatinib(260-340mg/m^2.d) or DAS(40-80 mg/m^2.d)+Chemotherapy	9	Chemotherapy	OS, EFS
3 Guo 2015 [24]	China	2008.10-2013.12	8(2-15)	D15 of remission induction	27	Imatinib(260-340mg/m^2.d) +Chemotherapy	26	Chemotherapy	EFS, ADR
4 Liu 2013 [25]	China	2003.10-2012.12	11(2-17)	D15 of remission induction	38	imatinib(260-340mg/m^2.d) + Chemotherapy	54	Chemotherapy	OS,EFS

Abbreviation: OS, overall survival; EFS, event-free survival; ADR, adverse drug reaction.

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Table 2. The quality assessment in cohort studies.

items	study	Zhang 2019 [22]	Wang 2016[23]	Guo 2015[24]	Liu 2013[25]
Selection	Representativeness of the exposed cohort	1	1	1	1
	Selection of the non exposed cohort	1	1	1	1
	Ascertainment of exposure	1	1	1	1
	Demonstration that outcome of interest was not present at start of study	1	1	1	1
Comparability	Comparability of cohorts on the basis of the design or analysis	2	1	1	1
Outcome	Assessment of outcome	1	1	1	1
	Was follow-up long enough for outcomes to occur	1	1	1	0
	Adequacy of follow up of cohorts	0	1	1	1
NOS score		8	8	8	7

Figure Legends

Figure1. PRISMA flow chart

This flow diagram illustrated the results of search and the process of screening and selecting studies for inclusion, and the reasons for exclusions in this review.

Figure 2. The quality assessment of randomized clinical trials.

This plot is created by the software of RevMan 5.3. It illustrated the quality of included randomized clinical trials with each of the judgement ('low risk', 'high risk' or 'unclear risk' of bias).

Figure 3. Meta-analysis of the effectiveness of TKIs on overall survival, TKI vs non-TKI.

The square data markers represent hazard ratio (HR); horizontal lines, the 95% CIs with marker size reflecting the statistical weight of the study using fixed-effects meta-analysis. A diamond data marker represents the overall HR and 95% CI for the outcome of interest.

Figure 4. Meta-analysis of the effectiveness of TKIs on overall survival, dasatinib vs imatinib. The square data markers represent hazard ratio (HR); horizontal lines, the 95% CIs with marker size reflecting the statistical weight of the study using fixed-effects meta-analysis. A

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diamond data marker represents the overall HR and 95% CI for the outcome of interest.

Figure 5. Meta-analysis of the effectiveness of TKIs on event-free survival, TKI vs non-TKI. The square data markers represent hazard ratio (HR); horizontal lines, the 95% CIs with marker size reflecting the statistical weight of the study using fixed-effects meta-analysis. A diamond data marker represents the overall HR and 95% CI for the outcome of interest.

Figure 6. Meta-analysis of the effectiveness of TKIs on event-free survival, dasatinib vs imatinib. The square data markers represent hazard ratio (HR); horizontal lines, the 95% CIs with marker size reflecting the statistical weight of the study using fixed-effects meta-analysis. A diamond data marker represents the overall HR and 95% CI for the outcome of interest.

Figure 7. Meta-analysis of the safety of TKIs on adverse drug reaction, TKI vs non-TKI. The square data markers represent risk ratio (RR); horizontal lines, the 95% CIs with marker size reflecting the statistical weight of the study using random-effects meta-analysis. A diamond data marker represents the overall RR and 95% CI for the outcome of interest.

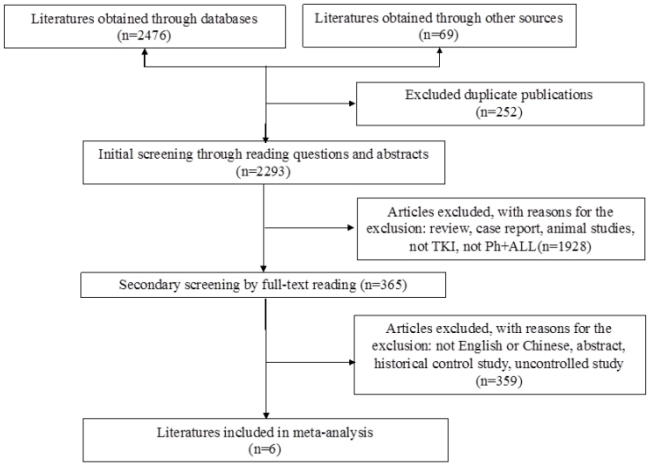
Figure 8. Meta-analysis of the safety of TKIs on adverse drug reaction, dasatinib vs imatinib. The square data markers represent risk ratio (RR); horizontal lines, the 95% CIs with marker size reflecting the statistical weight of the study using fixed-effects meta-analysis. A diamond data marker represents the overall RR and 95% CI for the outcome of interest.

Supporting Information

Appendix 1：Search strategies for PUBMED, CENTRAL and EMBASE.

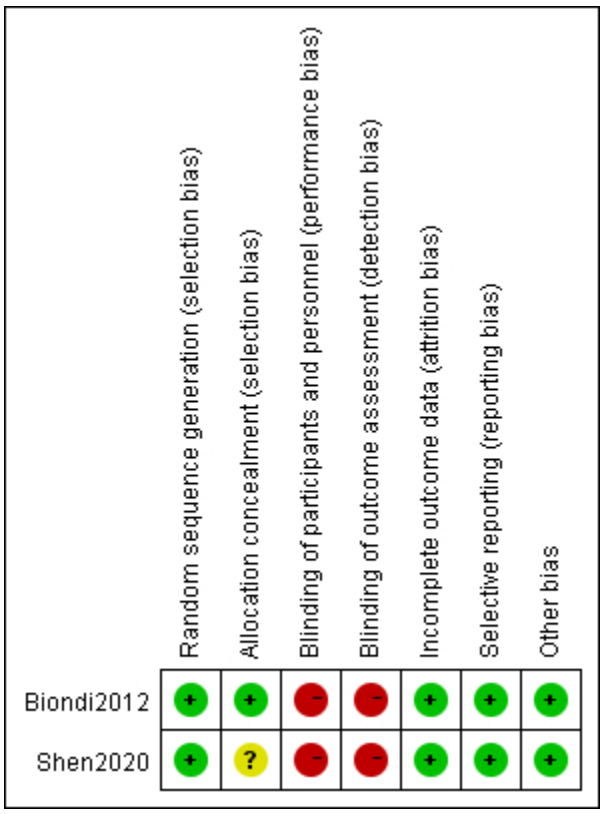
Appendix 2：Search strategies for CBM, CNKI, VIP and WANFANG.

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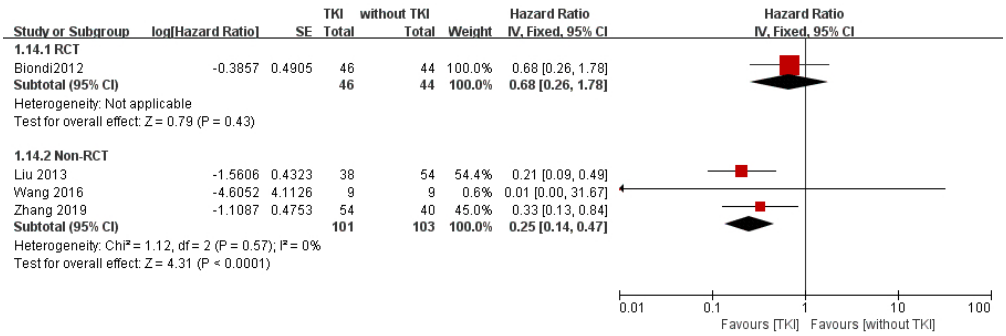
PRISMA flow chart

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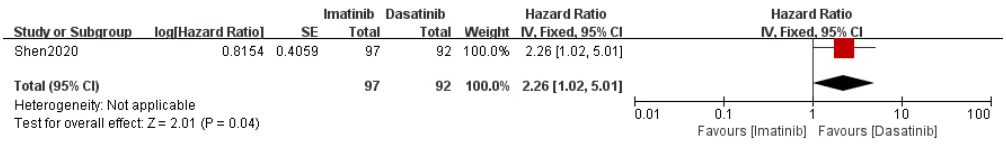
The quality assessment of randomized clinical trials.

105x141mm (72 x 72 DPI)



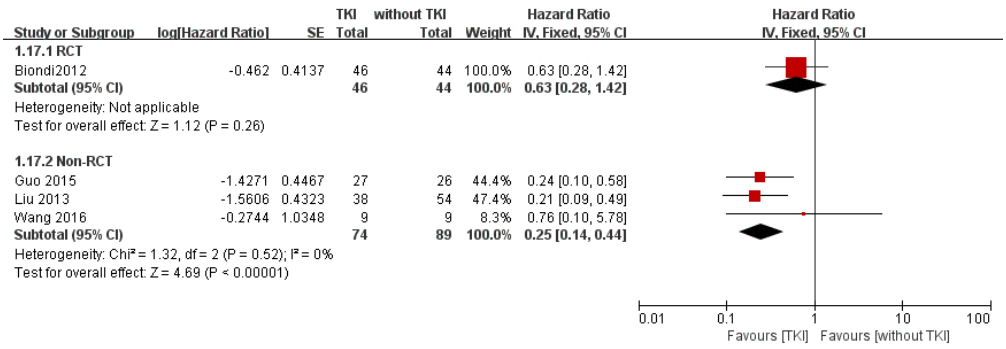
Meta-analysis of the effectiveness of TKIs on overall survival, TKI vs non-TKI.

318x107mm (72 x 72 DPI)



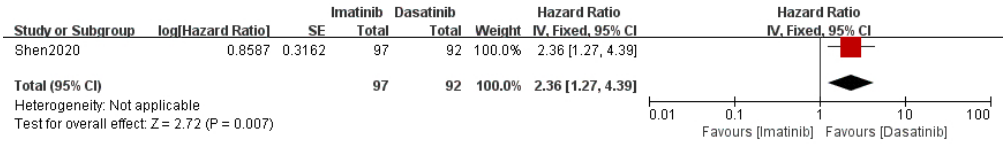
Meta-analysis of the effectiveness of TKIs on overall survival, dasatinib vs imatinib.

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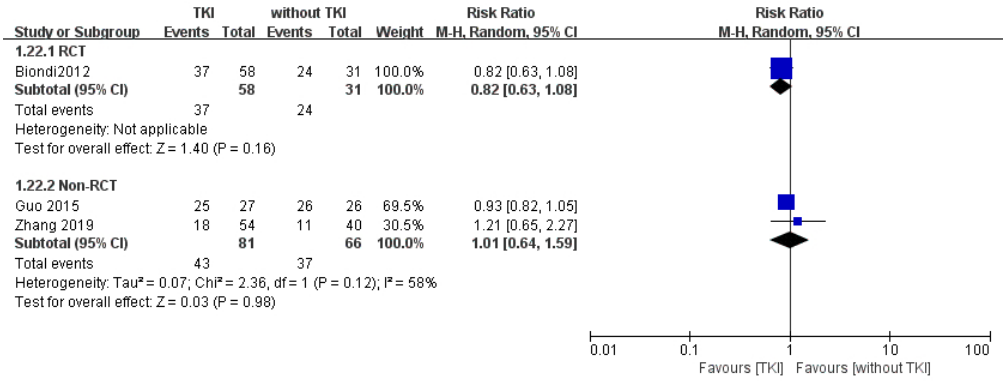
Meta-analysis of the effectiveness of TKIs on event-free survival, TKI vs non-TKI.

306x107mm (72 x 72 DPI)



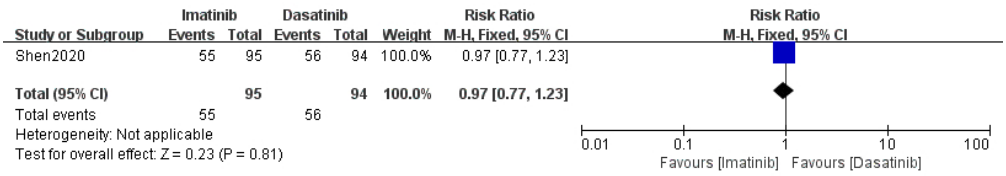
Meta-analysis of the effectiveness of TKIs on event-free survival, dasatinib vs imatinib.

305x45mm (72 x 72 DPI)



Meta-analysis of the safety of TKIs on adverse drug reaction, TKI vs non-TKI.

294x113mm (72 x 72 DPI)



Meta-analysis of the safety of TKIs on adverse drug reaction, dasatinib vs imatinib.

286x55mm (72 x 72 DPI)

Appendix 1: Search strategies for English databases

Databases	Search strategies
PubMed	#1"Philadelphia Chromosome"[Mesh]) OR "Ph Chromosome positive" OR "Philadelphia Chromosome positive" #2"Precursor Cell Lymphoblastic Leukemia-Lymphoma"[Mesh] OR "Acute Lymphoblastic Leukemia" OR "Acute Lymphocytic Leukemia" OR "Acute Lymphoid Leukemia" OR "Acute Lymphoblastic Lymphoma" #3"Imatinib Mesylate"[mesh] OR "imatinib"[mesh] OR "Imatinib" OR "Dasatinib" OR "nilotinib" OR " ponatinib" OR " tyrosine kinase inhibitor" OR "TKI" #4 #1 AND #2 AND #3
EMBASE	#1 " Philadelphia 1 chromosome"[thesaurus] OR " Philadelphia 1 chromosome positive" OR "Philadelphia Chromosome positive" #2 " acute lymphoblastic leukemia"[thesaurus] OR "Acute Lymphoblastic Leukemia" OR "Acute Lymphocytic Leukemia" OR "Acute Lymphoid Leukemia" OR "Acute Lymphoblastic Lymphoma" #3 " protein tyrosine kinase inhibitor"[thesaurus] OR "Imatinib" OR" Imatinib Mesylate"OR "Dasatinib" OR "nilotinib" OR " ponatinib" OR " tyrosine kinase inhibitor" OR "TKI" #3 #1 AND #2
Cochrane library	#1"Philadelphia Chromosome"[Mesh]) OR "Ph1 Chromosome positive" OR "Philadelphia Chromosome positive" #2"Precursor Cell Lymphoblastic Leukemia-Lymphoma"[Mesh] OR "Acute Lymphoblastic Leukemia" OR "Acute Lymphocytic Leukemia" OR "Acute Lymphoid Leukemia" OR "Acute Lymphoblastic Lymphoma" #3 "Imatinib Mesylate"[mesh] OR "imatinib"[mesh] OR "Imatinib" OR "Dasatinib" OR "nilotinib" OR " ponatinib" OR " tyrosine kinase inhibitor" OR "TKI" #4 #1 AND #2 AND #3

Appendix 2: Search strategies for Chinese databases

Databases	Search strategies
CBM	全部字段"费城染色体"and"急性淋巴细胞白血病"
CNKI	FT="费城染色体" and FT="TKI"+"酪氨酸激酶抑制剂"+"伊马替尼"+"尼洛替尼"+"达沙替尼"+"帕纳替尼"and FT="ALL"+"急性淋巴细胞白血病"+"急淋"
VIP	U="费城染色体"and"急性淋巴细胞白血病"
WANFANG	全部字段"费城染色体"and"急性淋巴细胞白血病"

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	2,3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3，PROSPERO No.CRD42018104107.
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	3,4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	3(Appendix 1)
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5,6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6,7(Figure 1)
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7(Table 1)
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7(Figure 2,Table 2)
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	NA
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	8,9(Figure 3-8)
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	8,9(Figure 3-8)
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	10-12
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	12
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	12-13
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	13

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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BMJ Open

Use of tyrosine kinase inhibitors for pediatric philadelphia chromosome-positive acute lymphoblastic leukemia: a systematic review and meta-analysis.

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-042814.R1
Article Type:	Original research
Date Submitted by the Author:	02-Oct-2020
Complete List of Authors:	Chen, Min; Sichuan University West China Second University Hospital, Department of Pharmacy zhu, yiping; Sichuan University West China Second University Hospital, Department of Pediatric Hematology and Oncology Lin, Yunzhu; Sichuan University West China Second University Hospital, pharmacy Tengwang, Tianzi; Sichuan University West China School of Pharmacy, pharmacy zhang, lingli; Sichuan University West China Second University Hospital, pharmacy
Primary Subject Heading:	Evidence based practice
Secondary Subject Heading:	Haematology (incl blood transfusion), Oncology, Paediatrics
Keywords:	Leukaemia < HAEMATOLOGY, Leukaemia < ONCOLOGY, Paediatric oncology < PAEDIATRICS, Clinical trials < THERAPEUTICS

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Use of tyrosine kinase inhibitors for pediatric philadelphia chromosome-positive acute lymphoblastic leukemia: a systematic review and meta-analysis.

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ABSTRACT

Objectives To investigate the effectiveness and safety of Tyrosine kinase inhibitors (TKIs) in the management of pediatric Philadelphia chromosome-positive acute lymphoblastic leukemia (Ph+ALL).

Design A systematic review and meta-analysis.

Data sources Electronic searches were conducted on CENTRAL, MEDLINE, EMBASE, SIOP, ASPHO, ASCO, ASH and four Chinese databases from inception to 8 March, 2020. Language of publications was restricted in English and Chinese.

Eligibility Criteria Prospective and retrospective comparative studies were included.

Data extraction and synthesis Two authors independently assessed and extracted data. Quality of studies was assessed by the Cochrane Collaboration's tool and Newcastle-Ottawa scale. Subgroup analysis was performed by comparing different types of TKIs. Quality of evidence was assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach.

Results Two randomized controlled trials (RCTs) and four cohort studies enrolling 536 patients were included. For RCTs, the pooled hazard ratio (HR) was 0.68(95%CI, 0.26-1.78) in overall survival (OS), 0.63(95%CI: 0.28-1.42) in event-free survival (EFS), respectively, comparing TKI arm with non-TKI arm for treatment of pediatric Ph+ALL. There was significant difference in OS and EFS between imatinib arm and dasatinib arm (HR: 2.26, 95%CI: 1.02-5.01; HR: 2.36; 95%CI: 1.27-4.39, respectively). For cohort studies, the pooled HR was 0.25(95%CI: 0.14-0.47) in OS, 0.25(95% CI: 0.12-0.56) in EFS, respectively, comparing TKI arm with non-TKI arm. There was no significance difference in adverse drug reaction (ADR) between TKI group and without TKI group (RR: 0.82, 95%CI: 0.63-1.08 in RCT; RR: 1.01, 95%CI:

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0.64-1.59 in cohort studies; respectively), and imatinib versus dasatinib (RR: 0.97, 95%CI: 0.77-1.23). The quality of evidence was rated as low for OS, EFS and ADR.

Conclusions The combination of TKIs with chemotherapy is likely to improve the OS and EFS rates in pediatric Ph+ALL, and dasatinib is superior than imatinib. Large sample size and prospective controlled studies are warranted.

Keywords: tyrosine kinase inhibitor; Ph+ALL; pediatric

PROSPERO registration number CRD42018104107

Strengths and limitations of this study

This study is the first systematic review and meta-analysis of the effectiveness and safety of TKIs in the management of pediatric Ph+ALL.

The present review included both prospective and retrospective comparative studies.

The Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach was used to evaluate the quality of the evidence.

This review was limited by the small number of studies and the heterogeneity of study design.

Introduction

Philadelphia chromosome-positive acute lymphoblastic leukemia (Ph+ALL), which occurs in approximately 3-5% of pediatric ALL patients, is recognized as a severe disease but leading to a dismal prognosis compared to those with philadelphia chromosome-negative ALL.[1]

In the past decades, intensive chemotherapy has been used in the treatment of pediatric Ph+ALL patients. [2,3] Later, hematopoietic stem cell transplant (HSCT) in first complete remission (CR1) was considered the treatment of choice due to the improved rate of overall survival (OS) and event-free survival (EFS) than intensive chemotherapy alone. However, the mortality rate associated with transplantation was almost 50%. [4,5] Recent studies suggested that tyrosine kinase inhibitors (TKIs) targeting BCR-ABL fusion protein combined with chemotherapy maybe an alternative effective therapy. [6] In adult patients with Ph+ALL, the use of TKIs in combination with chemotherapy were capable to increase the CR and HSCT rates and improved the early outcome. [7] In pediatric patients, the Children's Oncology Group (COG) trial AALL0031 had shown 80% 3-year EFS for Ph+ALL patients treated with intensive chemotherapy plus continuous 340mg/m^2 imatinib. The addition of imatinib

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to chemotherapy has similar outcomes to HSCT, especially in patients who responded favorably. [8] Long-term follow-up confirmed the favorable outcomes for these patients treated with imatinib plus intensive chemotherapy and HSCT seemed to be of no benefit. [9] However, the trial was observational and confirmatory results are not yet available from the European intergroup study on post induction treatment of ph+ALL with imatinib trial. [10]

Newer TKIs including dasatinib and nilotinib, were developed for those patients with resistance to imatinib. The resistance mechanisms are commonly associated with acquiring kinase domain mutations, reduced drug availability and activation of other signaling pathways such as the SRC family kinases. [11,12] Dasatinib is the more commonly used dual ABL/SRC kinase inhibitor. It can cross the blood-brain barrier to eradicate central nervous system (CNS) leukemia, and play a role in most cases of imatinib resistance. [13,14] For patients with BCR-ABL T315I mutation ph+ALL, the third-generation TKI ponatinib was reported as a valuable alternative treatment option. [15,16]

Incorporation of TKIs into frontline regimens for patients with newly diagnosed Ph+ ALL may be the current general consensus. However, this view may not be universally accepted. It will be helpful to analyze systematically to determine if it is truly beneficial to add TKIs to conventional chemotherapy. Therefore, our objective is to investigate the current available evidence on the effectiveness and safety of TKIs in the management of pediatric Ph+ALL.

Methods

This protocol was registered in the international PROSPERO register of systematic reviews: CRD42018104107.

Patient and Public Involvement

Patients and the public were not be involved in this review.

Search strategy

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We searched the Cochrane Central Register of Controlled Trials (CENTRAL) published in the Cochrane Library, MEDLINE in PubMed, EMBASE (Ovid), the Chinese Biomedical Literature Database (CBM), Chinese National Knowledge Infrastructure (CNKI), VIP Database for Chinese Technical Periodicals (VIP) and WANFANG for English and Chinese references. The full search strategies used in this study were detailed in online supplementary file 1. Retrieval time was from the inception of the database to 8 March, 2020.

The following societies of conference proceedings of annual meetings (from the inception of TKIs treatment till 8 March, 2020) were scanned：

1. International Society for Paediatric Oncology (SIOP);
2. American Society for Pediatric Hematology and Oncology (ASPHO);
3. American Society of Clinical Oncology (ASCO);
4. American Society of Hematology (ASH).

Inclusion and exclusion criteria

Study designs

There were limited randomized controlled trials (RCTs) in the treatment of pediatric patients with Ph+ALL, hence, we included retrospective and prospective cohort/case-control/randomized studies. Studies that did not include a comparative group of patients, duplicate studies, abstracts and historical control studies were excluded.

Participants

All patients aged 1-18 years with Ph positive status diagnosed by cytogenetic or molecular were included. Participants with ph-negative at diagnosis but positive at relapse, more than two types of malignant neoplasm, relapse or refractory Ph+ALL were excluded.

Interventions

TKIs alone, TKIs combined with other intervention

Comparators

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Treatment without TKIs

Outcomes

1. OS, defined as time from start of treatment to death from any cause;

2. EFS, defined as time from start of treatment to an event, such as remission failure, relapse, death from any cause, abandonment of treatment, second malignancy or to the date of last follow-up contact;

3. Adverse drug reaction (ADR)， collected and scored according to the National Cancer Institute Common Toxicity Criteria for Adverse Events (NCI-CTCAE) or other criterion by investigators and clinical research coordinators.

Data extraction

Literature search results were uploaded to EndNote X7. Two authors independently screened the title and abstract of all studies identified by the search strategy and obtained full articles for all potentially relevant trials. Disagreements were resolved by consensus between authors or achieved final resolution using a third-party arbitrator.

Two authors independently used a pre-designed data collection form to extract data from each study. The items included: characteristics of the trial (author, country, publication year, design etc), participants (age, sex etc), interventions (dose, regimen etc), outcomes and length of follow-up.

Risk of bias in individual studies

Two authors independently assessed the risk of bias using a standard form. For randomized trials, we used the domain-based evaluation recommended by the Cochrane Handbook for Systematic Reviews of Intervention to address the following domains: bias arising from the randomization process, bias due to deviations from intended interventions, bias due to missing outcome data, bias in measurement of the outcome, bias in selection of the reported result. [17] Plots of "Risk of Bias" assessment were created using Review Manager 5 (RevMan 5). For non-randomized trials, we

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used the Newcastle-Ottawa Scale (NOS) items that were categorized into three broad perspectives: the selection of the study groups, the comparability of the groups, and the ascertainment of either the exposure or outcome of interest for case-control or cohort studies respectively. [18] The maximum total score of NOS is 9 points.

Data analysis and synthesis

Strategy for data analysis

For dichotomous data, we used the risk ratio (RR) with 95% confidence interval (CI) as the effect measure. For time-to-event data, which take into account of the number and timing of events, we summarized and analyzed data using hazard ratio (HR) with their corresponding 95% CI. [17] We used Parmar's method if HR was not explicitly presented in the research. [19]

Assessment of heterogeneity

We quantified heterogeneity using the I^2 statistic, which illustrated the percentage of the variability in effect estimates resulting from heterogeneity. In the absence of significant heterogeneity (I^2 less than 50%), we used a fixed-effect model for the estimation of treatment effect. [17] Otherwise, we explored possible reasons for the occurrence of heterogeneity and used random-effects model.

Data synthesis

We entered data into the RevMan 5 software and made analyses according to the guidance provided in the Cochrane Handbook for Systematic Reviews of Interventions. [20] We pooled results only if both treatment groups were comparable, including the outcome definition. Otherwise, we presented a narrative summary. If different analysis methods (such as intention-to-treat analysis, as-treated analysis) are used in the evaluation of benefits between groups, we adopt the results of the intention-to-treat analysis to reduce bias.

Assessment of reporting biases

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If we include 10 or more trials, we use funnel plots to assess biases including publication bias and other reporting biases. If there are biases, the funnel plots may be asymmetrical.

Subgroup analysis

Considering that different types of TKIs might affect the outcomes, we conducted a subgroup analysis.

Grading the evidence

We used the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach to rate the quality of evidence. [21] In the GRADE approach, RCTs start as high-quality evidence. Five factors may lead to downgrading the quality of evidence. These factors include risk of bias, inconsistency, indirectness, imprecision and publication bias. Observational studies (including cohort and case-control studies) start as low-quality evidence. Three factors may lead to upgrading the quality of evidence, and these factors are large effect, dose response and all plausible residual confounding. The quality of the evide nce was rated as high, moderate, low or very low. We used GRADEpro software (GRADEpro 2011) to create a "Summary of findings" table for all three outcomes listed above.

Results

Study Selection

Our search strategy identified 2476 references from the electronic searches and 69 from other sources. 252 duplicate publications and 1928 articles were excluded after reviewing the title and abstract because they failed to meet the inclusion criteria such as type of article, study design, population or outcome of interest. 365 articles underwent full-text review and six literatures were included in the final data analysis. The PRISMA flow chart of bibliographic search was illustrated in Figure 1.

Characteristics of Included Studies

Six articles [10,22-26] (two prospective randomized open-label controlled trials and four retrospective cohort studies) were included in our analysis. Five studies [10,23-26] evaluated the efficacy and safety of TKIs in association with multidrug chemotherapy

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or chemotherapy alone in pediatric patients with Ph+ALL. Two of these studies [25,26] came from the same research institution. Among them, one study [25] retrospectively analyzed 53 cases between 2008 to 2013. The other [26] retrospectively analyzed 92 cases between 2003 to 2012. One RCT [22]compared the effect of dasatinib versus imatinib. Overall, 536 children aged 1 to 18 years with a confirmed diagnosis of Ph+ ALL contributed to the analysis. The chemotherapy regimen was modeled on Berlin-Frankfurt-Munster (BFM) high-risk arm, the Chinese Childhood Leukemia Group ALL (CCLG-ALL) high-risk arm and the Chinese Children's Cancer Group ALL (CCCG-ALL) intermediate-risk arm. TKI drugs include first generation imatinib and second generation dasatinib. Patients were prescribed oral imatinib (260-340mg/m^2.d) or dasatinib (40-80mg/m^2.d) on day 8, day 22 of remission induction or the end of induction. The exposure of TKIs ranged from 56 days to the end of therapy. Table 1 described the detailed characteristics of the studies included in this meta-analysis.

Risk of Bias Assessment in Included Studies

Figure 2 provided quality assessment of RCTs and Table 2 described quality assessment of non-RCTs. In the two prospective trials, stratified randomization was done centrally with web-based randomization system and both were open-label. Therefore, the random sequence generation was low risk while the blindness was high risk. All studies had low risk bias in incomplete outcome data, selective reporting and other issues. In four cohort studies, the NOS methodological score was 7-8 points. In the selection section, both exposed cohort and unexposed cohort were drawn from the same source of participant. Data was collected and retrospective reviewed from medical records. Outcome was not present at the beginning of all studies. In the comparability section, the three studies did not describe important confounding factors, except for TKI treatment between exposed and unexposed cohort among pediatric Ph+ALL. In the outcome section, one study was followed up for 2 years, which might not be long enough for outcomes to occur. One study did not describe the patient's loss of follow-up. Overall, the retrospective research has high quality NOS scores. Therefore, the risk of bias of RCTs was rated as moderate and the cohort studies were rated as low.

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TKI versus non-TKI

Overall Survival

Four studies (one RCT, three cohort studies) [10,23,24,26] with a total of 294 patients were included in the meta-analysis of OS for pediatric Ph+ALL patients. The pooled HR of OS in patients with TKI use was 0.68 (95% CI: 0.26-1.78, *P*=0.43) in RCT and 0.25 (95% CI: 0.14-0.47, *P*<0.05) in cohort studies, respectively. Figure 3 showed the forest plot of the included studies.

Event-free Survival

The primary outcome in one RCT [10] was disease-free survival (DFS) because the group only included good-risk patients in CR1 and poor-risk patients were not included. Therefore, in this meta-analysis, DFS is considered equivalent to EFS. Four studies (one RCT, three cohort studies) [10,24-26] reported the EFS. Because Liu et al [26] covered a longer period with larger number of patients, its results for EFS was included and those from Guo et al [25] was excluded. Finally, a total of 200 patients were included. The pooled HR of EFS in Ph+ALL patients using TKI was 0.63 (95%CI: 0.28-1.42, *P*=0.26) in RCT and 0.25 (95%CI: 0.12-0.56, *P*<0.05) in cohort studies, respectively. Figure 4 showed the forest plot of the included studies.

Adverse Drug Reaction

Three studies (one RCT, two cohort studies) [10,23,25] reported serious ADR in patients with Ph+ALL. Heterogeneity in the cohort studies was significant (I^2 =58%). We used a random-effect model for the estimation of treatment effect. In RCT, myelosuppression was the main cause of ADR. The proportions of patients with the most commonly reported serious ADR did not differ substantially between imatinib group and non-imatinib group (*P*=0.64). The most common serious ADR was infection, including fungal infection, localized infection and others. The pooled RR of serious ADR in patients with TKI use was 0.82 (95%CI: 0.63-1.08, *P*=0.16) in RCT. Result of the cohort studies was consistent with RCT. The pooled RR was 1.01 (95%CI: 0.64-1.59, *P*=0.12) (Figure 5).

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Different TKIs

One RCT [22] with a total of 189 Ph+ALL children compared the OS, EFS and ADR of oral imatinib at a daily dosage of 300mg/m^2 versus dasatinib at a daily dosage of 80mg/m^2, in combination with an intensive chemotherapy backbone. The HR of OS was 2.26 (95%CI: 1.02-5.01, *P*=0.04). The HR of EFS was 2.36 (95%CI: 1.27-4.39; *P*<0.05). There was no significant difference in the frequency of severe toxic effects between dasatinib and imatinib arms (RR: 0.97; 95%CI: 0.77-1.23, *P*=0.81). Infections constituted the most common serious adverse events. Approximately 5% of the patients in each arm died of fatal infections and 7% of patients in each treatment arm had disseminated fungal infections. The incidence of pleural effusion was 2% in the imatinib group and 4% in the dasatinib group. However, this difference was also not significant (*P*=0.44).

GRADE assessment

The evidence provided by the RCTs was low quality for both comparison of TKI based chemotherapy regimens versus chemotherapy alone, and the comparison of imatinib and dasatinib. The quality was downgraded by the risk of bias and imprecision due to high risk in blindness and wide 95% CI for effect estimates. Meanwhile, the evidence that the cohort studies provided to this review was of moderate quality. It was upgraded by the large magnitude of treatment effect. A summary of the GRADE assessment was shown in online supplementary file 2.

Discussion

TKIs have revolutionized the management of Ph+ALL patients. The national comprehensive cancer network (NCCN) has recommended use of TKIs based chemotherapy regimens for the adolescents and adults with Ph+ALL. [7] Warraich et al's systematic review comprising eighteen prospective and retrospective studies with 462 participates, showed that utilization of TKI (all generations) post allo-HSCT for adults patients in CR1 improved OS when given as prophylactic or pre-emptive regimen. [27] However, it is unclear whether this observed improvement of TKIs is

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significant in children. To our knowledge, our research is the first systematic review to investigate whether adding TKIs to conventional chemotherapy is indeed beneficial in pediatric Ph+ALL. Our research applied strict inclusion criteria and only those studies had comparative group of patients. Eventually, a meta-analysis was performed including one RCT and four non-RCTs with or without TKI treatment, and one RCT comparing dasatinib and imatinib in pediatric Ph+ALL.

TKI versus non-TKI

The included randomized trial EsPhALL 2004[10] demonstrated that imatinib in conjunction with intensive chemotherapy might be beneficial for treating children with Ph+ALL. The 4-year DFS in good-risk patients who received intensive chemotherapy and discontinuous post induction imatinib 300mg/m2.d increased by approximately 20% compared to patients receiving chemotherapy alone (75.2% versus 55.9%). The long-term follow up outcome showed 5-year DFS in the imatinib arm of 75.5% versus 61.4% in the no imatinib arm (P=0.20). [28] However, the benefit was not significantly. The pooled HR was 0.68 (95% CI: 0.26-1.78) in OS, 0.63 (95% CI: 0.28-1.42) in EFS, respectively, comparing TKIs arm with non-TKIs arm in combination with intensive chemotherapy. But the results were reversed by the meta-analysis including four high-quality cohort studies (NOS score ≥7). In these cohort studies, the pooled HR was 0.25 (95% CI: 0.14-0.47) in OS, 0.25 (95% CI: 0.12-0.56) in EFS, respectively, comparing TKIs arm with non-TKIs arm. The different results may be owing to the various regimen of timing and dosage of TKIs and the interference of HSCT during treatment.

In the EsPhALL 2004 study, the imatinib was used at a dose of 300mg/m^2 for 126 days after induction. In cohort studies, imatinib 260-340mg/m^2 was started after induction or earlier on day 15 of remission induction. Most of cohort studies have continued to use TKIs throughout chemotherapy unless severe infections, intolerance to TKIs or neutropenia with fever. Early and continuous exposure of imatinib may improve outcomes of OS and EFS. However, a prospective, open-label, single-arm

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clinical trial EsPhALL2010, the successor of EsPhALL 2004, administrated imatinib 300mg/m2.d continuously from day 15 of induction, displayed similar results to EsPhALL2004. [29]

The outcome of comparison between TKIs with chemotherapy versus HSCT treatment was limited in the current studies. The investigators from the COG reported the long-term outcome of pediatric Ph+ALL, showing the equivalent efficacy between the patients who received at least 280 continuous days of imatinib and those who underwent HSCT. [9] Two other retrospective studies indicated that there was no significant difference in OS or EFS between HSCT and chemotherapy with TKIs treatment, which was consistent with COG study. [30,31] In the included study EsPhALL 2004[10], about 80% of the patients had HSCT, 37 out of 46 in the imatinib group and 32 out of 44 in the non-imatinib group. This might lead to no statistical difference between the two groups of results in this RCT. However, the difference in the treatment outcomes of TKI combined with chemotherapy and HSCT for pediatric Ph + ALL have not been proven.

Different TKIs

In previous studies of imatinib, EFS was approximately 60% at 5 years, but resistance and relapse was common. Disappointingly, COG trial (AALL0622) for treatment of pediatric Ph+ ALL did not prove the superior of dasatinib over imatinib. OS and EFS were similar in the preceding trials using imatinib. [32] Another COG trial (AALL1122) using dasatinib 60mg/m^2.d is ongoing. [33] Now, Shen et al [22] reported different outcomes from the first randomized trial directly comparing dasatinib with imatinib. The 4-year OS rate in the dasatinib group was significantly higher than the imatinib group (88.4% versus 69.2%; *P*=0.04). The 4-year EFS rate for the 92 dasatinib-treated patients was 71.0%, significantly better than the 48.9% for the 97 imatinib-treated patients (*P*=0.005). In these 189 eligible patients, there was significant difference in OS and EFS between dasatinib arm and imatinib arm (HR: 2.26, 95%CI: 1.02-5.01;

HR: 2.36, 95%CI: 1.27-4.39; respectively). It is worth noting that the dose of dasatinib is higher than COG trials and longer follow-up is required to assess whether they represent significant improvement.

Adverse drug reaction

The dosing of imatinib (260-570mg/m^2.d) and dasatinib (50-110 mg/m^2.d) are well tolerated in children and adolescents with leukemia. [34,35] In the included studies, the daily dose of imatinib and dasatinib 260-340mg/m^2 and 40-80mg/m^2, did not increase the chemotherapy-related toxicity and side effects such as cytopenia and pleural effusions, suggesting that imatinib or dasatinib combined with intensive chemotherapy was well tolerated. [36] There was no significance difference in severe adverse drug reaction between TKI arm and without TKI arm (RR: 0.82, 95%CI: 0.63-1.08 in RCT; RR: 1.01, 95%CI: 0.64-1.59 in cohort studies; respectively). Similarly, no significant difference in the frequency of severe ADR was detected between dasatinib and imatinib arm (RR: 0.97, 95%CI: 0.77-1.23).

Quality of the evidence

The overall quality of the evidence assessed by the GRADE approach was low. Due to the rarity of Ph+ ALL in this age group and the dramatic improvement in survival with TKIs reported in observational researches, data from blinded controlled studies are very scarce. In addition, the sample size was small, making the 95%CIs wide. For this reason, we limited confidence in the estimation of effect values. Further studies are likely to have an important impact on our confidence in the estimated effect and may change our estimate.

Limitations

This meta-analysis has the following limitations. Firstly, the included studies are limited by the small study sample. In addition, HSCT is also a very important factor that needs to be taken into consideration. Even though the addition of the TKIs had improved the 5-year EFS and OS rates, but about 40% to 80% patients received transplant. Lastly, this is a meta-analysis of prospective and retrospective studies,

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which has its inherent limitations. Therefore, we did not combine the results of different types of studies.

Conclusions

Overall, low quality studies suggested that TKIs combined with intensive chemotherapy is likely to improve the OS and EFS rates in pediatric Ph+ALL, and the second-generation TKI dasatinib is superior than first-generation imatinib. The adverse drug reactions of TKIs could be tolerated.

This review highlights the need for future large sample size research for the use of TKIs in the treatment of pediatric Ph+ALL. Further prospective controlled studies are warranted to address remaining questions relating to the appropriate time to use TKIs during chemotherapy, as well as the role of HSCT in the management of these patients.

Data sharing statement

No additional data are available.

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arbitrated any disagreements and ensured that no errors occur during the review. All authors read, provided feedback and approved the final manuscript.

Competing interests None declared.

References

1. Schlieben S, Borkhardt A, Reinisch I, *et al*. Incidence and clinical outcome of children with BCR/ABL-positive acute lymphoblastic leukemia (ALL). A prospective RT-PCR study based on 673 patients enrolled in the German pediatric multicenter therapy trials ALL-BFM-90 and CoALL-05-92. *Leukemia* 1996;10(6):957-63.
2. Reiter A, Schrappe M, Ludwig WD, *et al*. Chemotherapy in 998 unselected childhood acute lymphoblastic leukemia patients. Results and conclusions of the multicenter trial ALL-BFM 86. *Blood* 1994;84(9):3122-33.
3. Ribeiro RC, Broniscer A, Rivera GK, *et al*. Philadelphia chromosome-positive acute lymphoblastic leukemia in children: durable responses to chemotherapy associated with low initial white blood cell counts. Leukemia 1997;11(9):1493-6.
4. Aricò M, Schrappe M, Hunger SP, *et al*. Clinical outcome of children with newly diagnosed Philadelphia chromosome-positive acute lymphoblastic leukemia treated between 1995 and 2005. *Journal of clinical oncology* 2010;28(31):4755-61.
5. Arico M, Valsecchi MG, Camitta B, *et al*. Outcome of treatment in children with Philadelphia chromosome-positive acute lymphoblastic leukemia. *N Engl J Med* 2000;342(14):998-1006.
6. Ravandi F. How I treat Philadelphia chromosome-positive acute lymphoblastic leukemia. *Blood* 2019;133(2):130-6.
7. NCCN. Clinical Practice Guidelines in Oncology. Acute Lymphoblastic Leukemia. Secondary Clinical Practice Guidelines in Oncology. Acute Lymphoblastic Leukemia 2018. https://www.nccn.org/professionals/physician_gls/default.aspx (accessed 10 September, 2020).
8. Schultz KR, Bowman WP, Aledo A, *et al*. Improved early event-free survival with imatinib in Philadelphia chromosome-Positive acute lymphoblastic leukemia: A Children's Oncology Group Study. *Journal of Clinical Oncology* 2009;27(31):5175-81.
9. Schultz KR, Carroll A, Heerema NA, *et al*. Long-term follow-up of imatinib in pediatric Philadelphia chromosome-positive acute lymphoblastic leukemia: Children's oncology group study AALL0031. *Leukemia* 2014;28(7):1467-71.
10. Biondi A, Schrappe M, De Lorenzo P, *et al*. Imatinib after induction for treatment of children and adolescents with Philadelphia-chromosome-positive acute lymphoblastic leukaemia (EsPhALL): a randomised, open-label, intergroup study. *The lancet oncology* 2012;13(9):936-45.
11. Brave M, Goodman V, Kaminskas E, *et al*. Sprycel for chronic myeloid leukemia and Philadelphia chromosome-positive acute lymphoblastic leukemia resistant to

For peer review only

or intolerant of imatinib mesylate. *Clin Cancer Res* 2008;14(2):352-9.
12. Kantarjian H, Shah NP, Hochhaus A, *et al*. Dasatinib versus imatinib in newly diagnosed chronic-phase chronic myeloid leukemia. *N Engl J Med* 2010;362(24):2260-70.
13. O'Hare T, Walters DK, Stoffregen EP, *et al*. In vitro activity of Bcr-Abl inhibitors AMN107 and BMS-354825 against clinically relevant imatinib-resistant Abl kinase domain mutants. *Cancer Res* 2005;65(11):4500-5.
14. Porkka K, Koskenvesa P, Lundan T, *et al*. Dasatinib crosses the blood-brain barrier and is an efficient therapy for central nervous system Philadelphia chromosome-positive leukemia. *Blood* 2008;112(4):1005-12.
15. Cortes JE, Kantarjian H, Shah NP, *et al*. Ponatinib in refractory Philadelphia chromosome-positive leukemias. *N Engl J Med* 2012;367(22):2075-88.
16. Nicolini FE, Basak GW, Kim DW, *et al*. Overall survival with ponatinib versus allogeneic stem cell transplantation in Philadelphia chromosome-positive leukemias with the T315I mutation. *Cancer* 2017;123(15):2875-80.
17. Higgins J GS. Cochrane handbook for systematic reviews of interventions. . Chichester, UK: The Cochrane Library, John Wiley &Sons https://methodscochraneorg/risk-bias-20-tool (accessed10 September, 2020).
18. Wells GA SB, O'Connell D, *et al*. The Newcastle-Ottawa Scale (NOS) for assessing the quality if nonrandomized studies in meta-analyses. http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp (accessed10 September, 2020).
19. Parmar MK, Torri V, Stewart L. Extracting summary statistics to perform meta-analyses of the published literature for survival endpoints. *Stat Med* 1998;17(24):2815-34.
20. Review Manager(RevMan)[Computer program] 2014. Version 5.3.Copenhagen: The Nordic Cochrane Centre, The Cochrane Collaboration.
21. Guyatt G, Oxman AD, Akl EA, *et al*. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *J Clin Epidemiol*. 2011;64(4):383-94.
22. Shen Sh, Chen XJ, Cai J, *et al*. Effect of Dasatinib vs Imatinib in the Treatment of Pediatric Philadelphia Chromosome-Positive Acute Lymphoblastic Leukemia: A Randomized Clinical Trial. *JAMA Oncology* 2020;6(3):358-66.
23. Zhang J, Li ZF, Guo J *et al*. Clinical Observation of Inductive Treatment of Childhood Acute Lymphoblastic Leukemia with Imatinib Mesylate Combined with VDLD Regimen. *Clinical Misdiagnosis & Mistherapy* 2019;32(9):26-30.
24. Wang CC, Jin RM. Clinical Analysis of 19 BCR-ABL Positive Childhood Acute Lymphoblastic Leukemia. Master dissertation. Huazhong University of Science and Technology. 2016
25. Guo Y, Liu TF, Ruan M *et al*. Efficacy and safety of imatinib for the treatment of Philadelphia chromosomepositive acute lymphoblastic leukemia in children. *Chin J Contemp Pediatr* 2015;17(8):819-24.
26. Liu TF, Zhu XF. Clinical and biological characteristics, therapeutic effects and prognostic factors of 92 children with Ph chromosome positive acute

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lymphoblastic leukemia. Doctor dissertation. Peking Union Medical College. 2013.

27. Warraich Z, Tenneti P, Thai T, *et al*. Relapse prevention with tyrosine kinase inhibitors after allogeneic transplantation for philadelphia chromosome-positive acute lymphoblast leukemia: a systematic review. Biol Blood Marrow Transplant. 2020;26(3):e55-64.
28. Biondi A, Cario G, De Lorenzo P, *et al*. Long-term follow up of pediatric Philadelphia positive acute lymphoblastic leukemia treated with the EsPhALL2004 study: high white blood cell count at diagnosis is the strongest prognostic factor. *Haematologica* 2019;104(1):e13-16.
29. Biondi A, Gandemer V, De Lorenzo P, *et al*. Imatinib treatment of paediatric Philadelphia chromosome-positive acute lymphoblastic leukaemia (EsPhALL2010): a prospective, intergroup, open-label, single-arm clinical trial. *Lancet Haematology* 2018;5(12):641-52.
30. Salami K, Alkayed K, Halalsheh H, *et al*. Hematopoietic stem cell transplant versus chemotherapy plus tyrosine kinase inhibitor in the treatment of pediatric Philadelphia chromosome-positive acute lymphoblastic leukemia (ALL). *Hematology/ Oncology and Stem Cell Therapy* 2013;6(1):34-41.
31. Gao Y, Guo Y, Hu SY, *et al*. Philadelphia chromosome-positive acute lymphoblastic leukemia in China: a retrospective study from the Chinese Childhood Cancer Group. *Leukemia & lymphoma* 2016;57(11):2696-8.
32. Slayton WB, Schultz KR, Kairalla JA, *et al*. Dasatinib Plus Intensive Chemotherapy in Children, Adolescents, and Young Adults With Philadelphia Chromosome-Positive Acute Lymphoblastic Leukemia: Results of Children's Oncology Group Trial AALL0622. *Journal of clinical oncology* 2018;36(22):2306-14.
33. ClinicalTrials.gov. Pediatric philadelphia positive acute lymphoblastic leukemia. Available at: https://www.clinicaltrials.gov/ct2/show/NCT01460160?term=AALL1122&draw=2&rank=1. Accessed 10July, 2020.
34. Champagne MA, Capdeville R, Krailo M, *et al*. Imatinib mesylate (STI571) for treatment of children with Philadelphia chromosome-positive leukemia: results from a Children's Oncology Group phase 1 study. *Blood* 2004;104(9):2655-60.
35. Aplenc R, Blaney SM, Strauss LC, *et al*. Pediatric phase I trial and pharmacokinetic study of dasatinib: a report from the children's oncology group phase I consortium. *Journal of clinical oncology* 2011;29(7):839-44.
36. Lindauer M HA. Dasatinib. *Recent Results Cancer Res* 2018;212:29-68.

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Table 1. Main characteristics of the studies included in this meta-analysis.

	Study/Year	Setting	Time	Population (years)	TKI start time	Intervention n1	drug	Comparison n2	drug	Outcomes
Randomized controlled trials										
1	Biondi 2012 [10]	Europe	2004.1-2009.12	1-18	post-induction, good risk	46	Imatinib(300mg/m^2.d) + Chemotherapy	44	Chemotherapy	OS,EFS, ADR
2	Shen 2020 [22]	China	2015.1-2018.9	7.8(0-18)	After diagnosis, usually on D8 of remission induction	97	Imatinib(300mg/m^2.d) + Chemotherapy	92	Dasatinib(80 mg/m^2.d)+ Chemotherapy	OS,EFS
Cohort studies										
1	Zhang 2019 [23]	China	2008.1-2012.12	8.26(2-12)	post-induction	54	Imatinib(0.6g/d)+ Chemotherapy	40	Chemotherapy	OS, ADR
2	Wang 2016 [24]	China	2008.10-2016.1	9(3-15)	Post-induction	9	Imatinib(260-340mg/m^2.d) or DAS(40-80 mg/m^2.d)+Chemotherapy	9	Chemotherapy	OS, EFS
3	Guo 2015 [25]	China	2008.10-2013.12	8(2-15)	D15 of remission induction	27	Imatinib(260-340mg/m^2.d)+Chemotherapy	26	Chemotherapy	EFS, ADR
4	Liu 2013 [26]	China	2003.10-2012.12	11(2-17)	D15 of remission induction	38	imatinib(260-340mg/m^2.d) + Chemotherapy	54	Chemotherapy	OS,EFS

Abbreviation: OS, overall survival; EFS, event-free survival; ADR, adverse drug reaction.

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Table 2. The quality assessment in cohort studies.

items	study	Zhang 2019 [23]	Wang 2016[24]	Guo 2015[25]	Liu 2013[26]
Selection	Representativeness of the exposed cohort	1	1	1	1
	Selection of the non exposed cohort	1	1	1	1
	Ascertainment of exposure	1	1	1	1
	Demonstration that outcome of interest was not present at start of study	1	1	1	1
Comparability	Comparability of cohorts on the basis of the design or analysis	2	1	1	1
Outcome	Assessment of outcome	1	1	1	1
	Was follow-up long enough for outcomes to occur	1	1	1	0
	Adequacy of follow up of cohorts	0	1	1	1
NOS score		8	8	8	7

Figure Legends

Figure1. PRISMA flow chart

This flow diagram illustrated the results of search and the process of screening and selecting studies for inclusion, and the reasons for exclusions in this review.

Figure 2. The quality assessment of randomized clinical trials.

This plot is created by the software of RevMan 5.3. It illustrated the quality of included randomized clinical trials with each of the judgement ('low risk', 'high risk' or 'unclear risk' of bias).

Figure 3. Meta-analysis of the effectiveness of TKIs on overall survival, TKI vs non-TKI.

The square data markers represent hazard ratio (HR); horizontal lines, the 95% CIs with marker size reflecting the statistical weight of the study using fixed-effects meta-analysis. A diamond data marker represents the overall HR and 95% CI for the outcome of interest.

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Figure 4. Meta-analysis of the effectiveness of TKIs on event-free survival, TKI vs non-TKI.
The square data markers represent hazard ratio (HR); horizontal lines, the 95% CIs with marker size reflecting the statistical weight of the study using fixed-effects meta-analysis. A diamond data marker represents the overall HR and 95% CI for the outcome of interest.

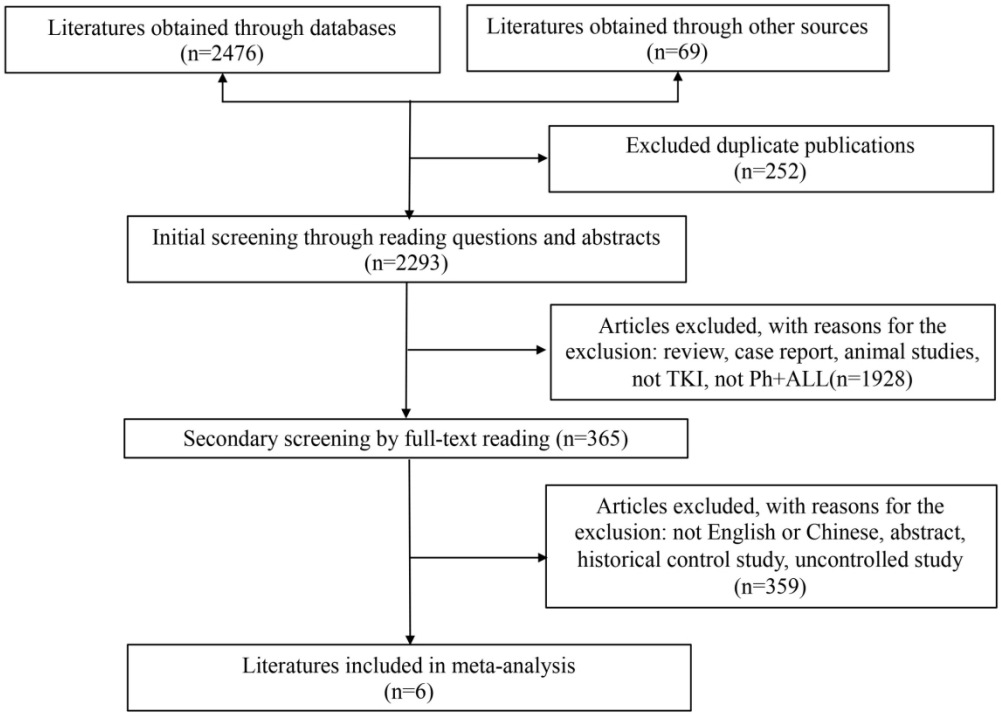
Figure 5. Meta-analysis of the safety of TKIs on adverse drug reaction, TKI vs non-TKI.
The square data markers represent risk ratio (RR); horizontal lines, the 95% CIs with marker size reflecting the statistical weight of the study using random-effects meta-analysis. A diamond data marker represents the overall RR and 95% CI for the outcome of interest.

Supporting Information

supplementary file 1：Search strategies for English and Chinese databases.

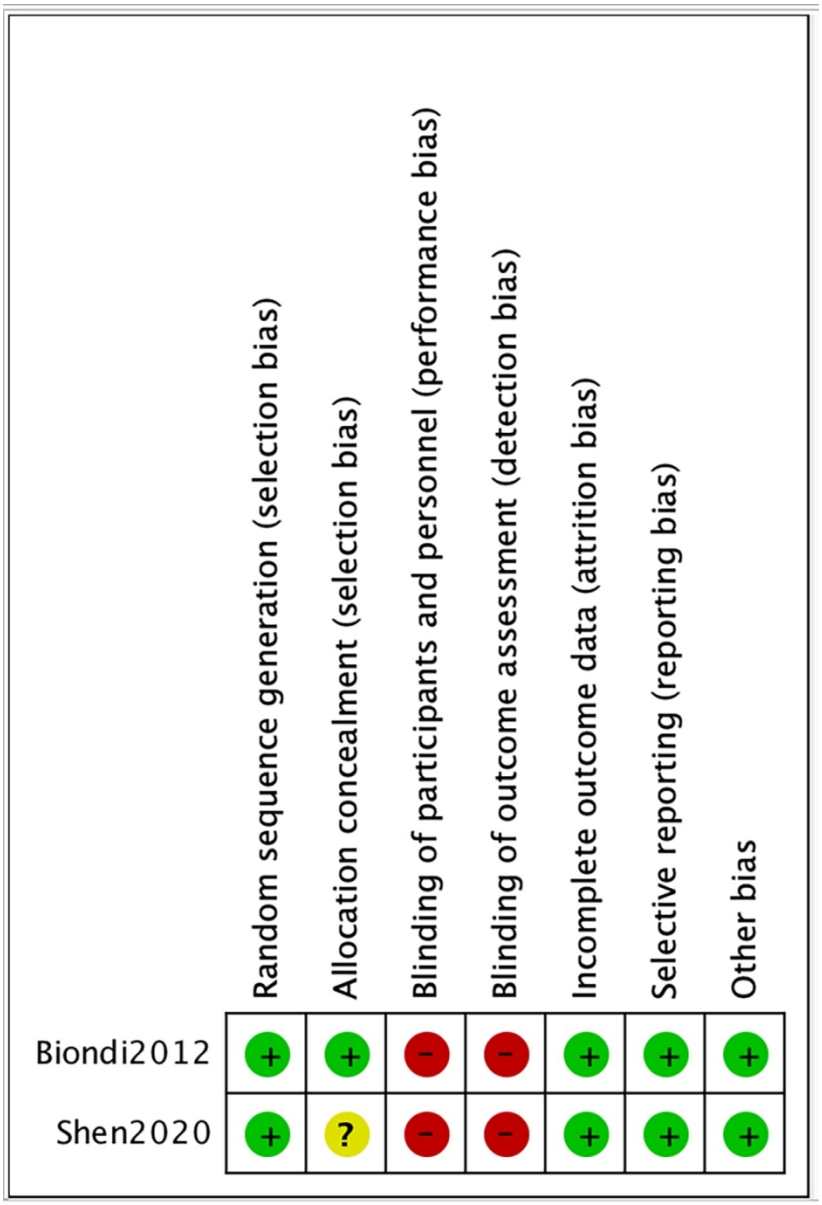
supplementary file 2: Summary of findings for the outcomes.

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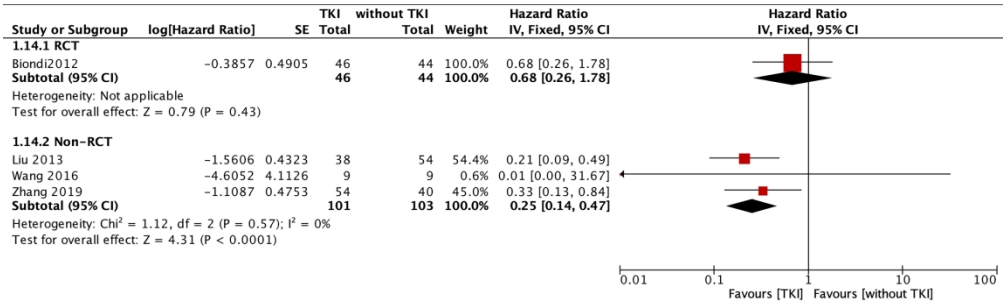
PRISMA flow chart

89x66mm (600 x 600 DPI)



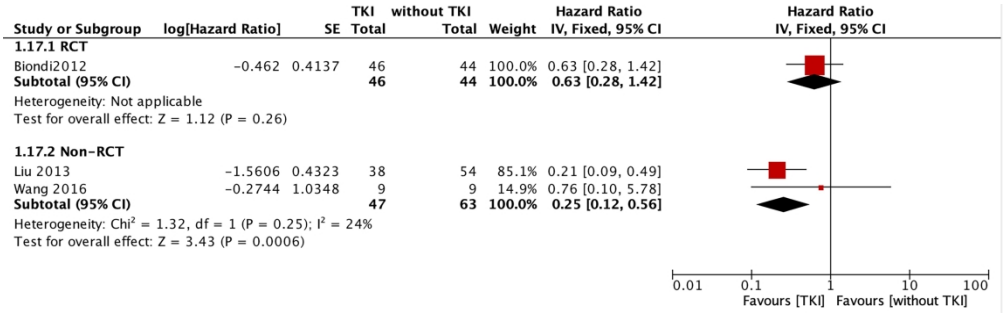
The quality assessment of randomized clinical trials.

108x159mm (300 x 300 DPI)



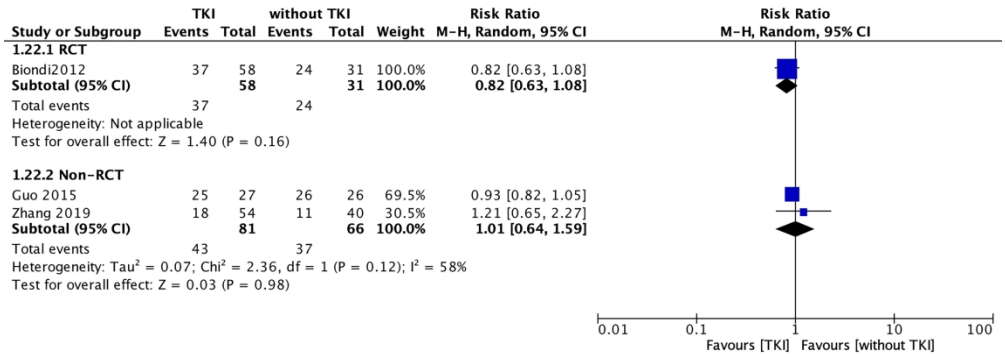
Meta-analysis of the effectiveness of TKIs on overall survival, TKI vs non-TKI.

352x117mm (300 x 300 DPI)



Meta-analysis of the effectiveness of TKIs on event-free survival, TKI vs non-TKI.

321x99mm (300 x 300 DPI)



Meta-analysis of the safety of TKIs on adverse drug reaction, TKI vs non-TKI.

315x118mm (300 x 300 DPI)

Appendix 1: Search strategies for English databases

Databases	Search strategies
PubMed	#1"Philadelphia Chromosome"[Mesh]) OR "Ph Chromosome positive" OR "Philadelphia Chromosome positive" #2"Precursor Cell Lymphoblastic Leukemia-Lymphoma"[Mesh] OR "Acute Lymphoblastic Leukemia" OR "Acute Lymphocytic Leukemia" OR "Acute Lymphoid Leukemia" OR "Acute Lymphoblastic Lymphoma" #3"Imatinib Mesylate"[mesh] OR "imatinib"[mesh] OR "Imatinib" OR "Dasatinib" OR "nilotinib" OR " ponatinib" OR " tyrosine kinase inhibitor" OR "TKI" #4 #1 AND #2 AND #3
EMBASE	#1 " Philadelphia 1 chromosome"[thesaurus] OR " Philadelphia 1 chromosome positive" OR "Philadelphia Chromosome positive" #2 " acute lymphoblastic leukemia"[thesaurus] OR "Acute Lymphoblastic Leukemia" OR "Acute Lymphocytic Leukemia" OR "Acute Lymphoid Leukemia" OR "Acute Lymphoblastic Lymphoma" #3 " protein tyrosine kinase inhibitor"[thesaurus] OR "Imatinib" OR" Imatinib Mesylate"OR "Dasatinib" OR "nilotinib" OR " ponatinib" OR " tyrosine kinase inhibitor" OR "TKI" #3 #1 AND #2
Cochrane library	#1"Philadelphia Chromosome"[Mesh]) OR "Ph1 Chromosome positive" OR "Philadelphia Chromosome positive" #2"Precursor Cell Lymphoblastic Leukemia-Lymphoma"[Mesh] OR "Acute Lymphoblastic Leukemia" OR "Acute Lymphocytic Leukemia" OR "Acute Lymphoid Leukemia" OR "Acute Lymphoblastic Lymphoma" #3 "Imatinib Mesylate"[mesh] OR "imatinib"[mesh] OR "Imatinib" OR "Dasatinib" OR "nilotinib" OR " ponatinib" OR " tyrosine kinase inhibitor" OR "TKI" #4 #1 AND #2 AND #3

Appendix 2: Search strategies for Chinese databases

Databases	Search strategies
CBM	全部字段"费城染色体"and"急性淋巴细胞白血病"
CNKI	FT="费城染色体" and FT="TKI"+"酪氨酸激酶抑制剂"+"伊马替尼"+"尼洛替尼"+"达沙替尼"+"帕纳替尼"and FT="ALL"+"急性淋巴细胞白血病"+"急淋"
VIP	U="费城染色体"and"急性淋巴细胞白血病"
WANFANG	全部字段"费城染色体"and"急性淋巴细胞白血病"

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Appendix 3: Summary of findings for the main outcomes.

TKI versus Non-TKI for pediatric Ph+ALL in randomized controlled trial

Patient or population: patients with pediatric Ph+ALL

Intervention: TKI with chemotherapy

Comparison: chemotherapy

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No of Participants (studies)	Quality of the evidence (GRADE)	Comments
	Assumed risk chemotherapy	Corresponding risk TKI with chemotherapy				
overall survival Follow-up: mean 3.1 years	**727 per 1000**	**587 per 1000** (287 to 901)	**HR 0.68** (0.26 to 1.78)	90 (1 study)	⊕⊕⊖⊖ **low**[1,2,3]	
event free survival Follow-up: mean 3.1 years	**614 per 1000**	**451 per 1000** (234 to 741)	**HR 0.63** (0.28 to 1.42)	90 (1 study)	⊕⊕⊖⊖ **low**[1,2,3]	
adverse drug reaction Follow-up: mean 3.1 years	**774 per 1000**	**635 per 1000** (488 to 836)	**RR 0.82** (0.63 to 1.08)	89 (1 study)	⊕⊕⊖⊖ **low**[1,2,4]	

*The basis for the **assumed risk** (e.g. the median control group risk across studies) is provided in footnotes. The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: Confidence interval; **RR:** Risk ratio; **HR:** Hazard ratio;

GRADE Working Group grades of evidence

High quality: Further research is very unlikely to change our confidence in the estimate of effect.

Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.

Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.

Very low quality: We are very uncertain about the estimate.

[1] blindness was high risk

[2] not applicable

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[3] the 95% CI included appreciable benefit or harm

[4] the 95%CI for effect estimates were wide

TKI versus Non-TKI for pediatric Ph+ALL in cohort studies

Patient or population: patients with pediatric Ph+ALL

Intervention: TKIs

Comparison: Non-TKIs

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No of Participants (studies)	Quality of the evidence (GRADE)	Comments
	Assumed risk **Non-TKIs**	Corresponding risk **TKIs**				
overall survival	**340 per 1000**	**99 per 1000** (56 to 177)	**HR 0.25** (0.14 to 0.47)	204 (3 studies)	⊕⊕⊕⊖ **moderate**[1]	
event free survival	**95 per 1000**	**25 per 1000** (12 to 55)	**HR 0.25** (0.12 to 0.56)	110 (2 studies)	⊕⊕⊕⊖ **moderate**[1]	
adverse drug reaction	**561 per 1000**	**566 per 1000** (359 to 891)	**RR 1.01** (0.64 to 1.59)	147 (2 studies)	⊕⊕⊕⊖ **moderate**[1]	

*The basis for the **assumed risk** (e.g. the median control group risk across studies) is provided in footnotes. The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: Confidence interval; **RR:** Risk ratio; **HR:** Hazard ratio;

GRADE Working Group grades of evidence

High quality: Further research is very unlikely to change our confidence in the estimate of effect.

Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.

Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.

Very low quality: We are very uncertain about the estimate.

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[1] the magnitude of the treatment effect is large

imatinib versus dasatinib for pediatric Ph+ALL in randomized controlled trial

Patient or population: patients with Ph+ALL

Intervention: imatinib

Comparison: dasatinib

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No of Participants (studies)	Quality of the evidence (GRADE)	Comments
	Assumed risk **Dasatinib**	Corresponding risk **Imatinib**				
overall survival Follow-up: mean 26.4 months	**880 per 1000**	**992 per 1000** (885 to 1000)	**HR 2.26** (1.02 to 5.01)	189 (1 study)	⊕⊕⊖⊖ **low**[1,2,3]	
event free survival Follow-up: mean 26.4 months	**707 per 1000**	**945 per 1000** (789 to 995)	**HR 2.36** (1.27 to 4.39)	189 (1 study)	⊕⊕⊖⊖ **low**[1,2,3]	
adverse drug reaction Follow-up: mean 26.4 months	**596 per 1000**	**578 per 1000** (459 to 733)	**RR 0.97** (0.77 to 1.23)	189 (1 study)	⊕⊕⊖⊖ **low**[1,2,4]	

*The basis for the **assumed risk** (e.g. the median control group risk across studies) is provided in footnotes. The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: Confidence interval; **RR:** Risk ratio; **HR:** Hazard ratio;

GRADE Working Group grades of evidence

High quality: Further research is very unlikely to change our confidence in the estimate of effect.

Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.

Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.

Very low quality: We are very uncertain about the estimate.

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[1] the blindness was high risk

[2] not applicable

[3] the 95% CI includes appreciable benefit or harm

[4] the 95%CI for effect estimates were wide

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	2,3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3，PROSPERO No.CRD42018104107.
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	3,4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	3(Appendix 1,2)
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5,6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7(Figure 1)
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7,8(Table 1)
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8(Figure 2,Table 2)
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	NA
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	9,10(Figure 3-5)
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	9,10(Figure 3-5)
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	10-13
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	13
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	14

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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